

HAT-P-54b

R-1015

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-54_260116 · created 2026-07-30T03:43:33+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461056.7231 +/- 0.0035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.158 +/- 0.022
Transit depth (Rp/Rs)^2	2.51 +/- 0.7 [%]
Orbital Inclination (inc)	86.99 +/- 1.52
Airmass coefficient 1 (a1)	1.218 +/- 0.029
Airmass coefficient 2 (a2)	-0.178 +/- 0.033
Scatter in the residuals of the lightcurve fit is	2.39 %
Best Comparison Star	None
Optimal Aperture	2.96
Optimal Annulus	6.57
Transit Duration (day)	0.067 +/- 0.0067

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.158 ± 0.022 vs 0.1572 (deviation 0.03σ)
Duration fit vs published	0.067 d vs 0.075 d (deviation 1.11σ)
Mid-time O–C	3.4 min
Depth SNR	6.7
Residual scatter	2.39 %

Light curve

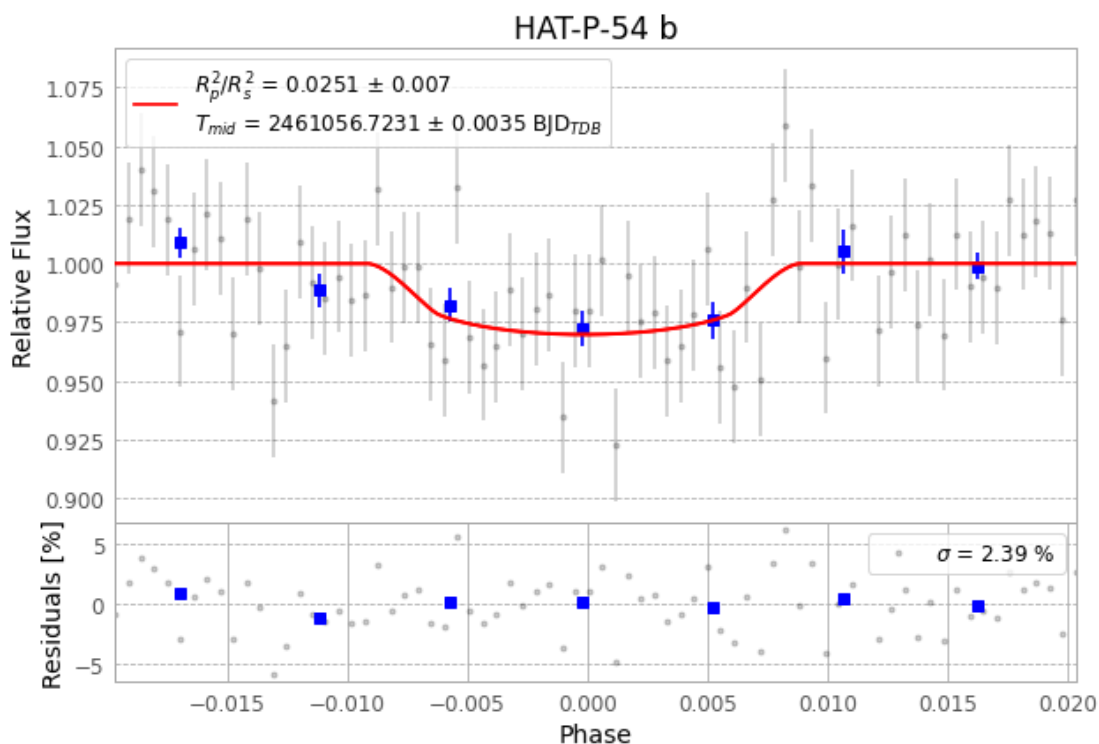


Figure 1 — FinalLightCurve_HAT-P-54 b_2026-01-15.png (SHA-256 a1d2ca97d54af6f6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [415, 270], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5baaad1455ddac03...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

74 FITS frames (49 MB), HATP-54260116032417.FITS → HATP-54260116070315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-260116.log (637a91796a90...), FinalLightCurve_HAT-P-54 b_2026-01-15.png (a1d2ca97d54a...), FinalLightCurve_HAT-P-54 b_2026-01-15.csv (e1f4a507f425...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 03:43Z.